

复赛提交说明 | 开放探索赛题

Semi-Final Submission Instructions | Open Exploration Sub-track

一、提交物总览：需提交 6 项材料

I. Overview of Deliverables: 6 Required Submission Items

	#	交付物 Deliverable	形式 Format	必交 Mandatory
非代码类材料 Non-Code Materials	1	方案说明 PPT Solution Presentation PPT	讲清问题定义、环境设计、探索过程与发现信号 Clarify problem definition, environment design, exploration process and discovered signals	必交 Yes
	2	科学发现与环境定义报告 Scientific Finding & Environment Definition Report	初赛问题定义文档的加强版，见第四节五个必答问题 Enhanced version of the preliminary-round problem definition document; see Section IV for five mandatory questions	必交 Yes

代码类材料 Code Materials	1	最小可运行探索环境 Minimal Operable Exploration Environment	「三件套」之一。他人能装能跑的环境本体 Component I of the “three-component set” . An environment that others can install and run	必交 Yes
	2	完整探索日志 Complete Exploration Log	「三件套」之二。至少一次完整运行记录，JSON / JSONL 格式 Component II of the “three-component set” . At least one complete execution record in JSON / JSONL format	必交 Yes
	3	参照系设计 Reference Benchmark Design	「三件套」之三。随机探索、平凡解等最小参照系及其结果 Component III of the “three-component set” . Minimal reference benchmark including random exploration, trivial solution and corresponding outputs	必交 Yes
	4	开源代码仓库 + README 与复现说明 Open-source Code Repository +	公开可访问链接，含一键 smoke test Publicly accessible link with one-click smoke-test support	必交 Yes

		README & Reproducibility Instructions		
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提交物分为**代码类**和**非代码类**,

提交材料需支持第三方理解、运行、审计并复核科学发现过程。

Submission deliverables are divided into **non-code** and **code** categories.

Submitted materials shall enable third-party parties to understand, execute, audit and review the scientific-discovery workflow.

- 代码类包括最小可运行探索环境、开源代码仓库、README、复现说明和必要配置；
- 非代码类包括科学发现与环境定义报告、完整探索日志、参照系结果、数据 / Benchmark 说明、结果证据和 PPT。
- Code materials include the minimal operable exploration environment, open-source code repository, README, reproducibility instructions and required configuration files.
- Non-code materials include the Scientific Finding & Environment Definition Report, complete exploration logs, reference benchmark outputs, data / Benchmark documentation, evidence for results, and presentation PPT.

二、提交方式

II. Submission Procedures

项 Item	内容 Content
压缩包命名	OPENxxx_作品名称_submission.zip (报告与日志)，仓库另附链接

Compressed-Package Naming	OPENxxx_SubmissionTitle_submission.zip (for reports and logs); repository link shall be provided separately
提交渠道 Submission Channel	官网 Official Website
截止时间 Deadline	2026 年 9 月 3 日（提交窗口 8 月 28 日—9 月 3 日 18:00） Sep. 3, 2026 (Submission Window: Aug. 28 – 18:00, Sep. 3)
仓库要求	固定一个 release / tag / commit hash ，避免评审期间仓库持续变化 Freeze a release / tag / commit hash to avoid ongoing repository modifications during evaluation

三、非代码类材料

III. Non-Code Materials

1. 方案说明 PPT

1. Solution Presentation PPT

复赛阶段设有线上答辩环节，PPT 需随作品一并提交，既作为评审通读材料，也作为答辩用材料。

An online defense session is scheduled for the semi-final round. **The PPT shall be submitted alongside your project**; it serves both as reading material for judges and presentation slides for defense.

PPT 至少说明以下内容：

The PPT shall cover at least the following items:

#	内容 Content	说明 Description
1	问题定义 Problem Definition	看见了什么问题，为什么 AI 可以介入，边界在哪 State the targeted problem, explain why AI can contribute, and clarify its boundaries
2	探索环境设计 Exploration Environment Design	固定什么、可探索什么、反馈机制是什么 Define fixed components, explorable components, and feedback mechanisms
3	探索过程与发现信号 Exploration Process & Discovered Signals	正结果、负结果、异常或问题修正各意味着什么；参照系对比 Interpret positive results, negative results, anomalies or problem revisions; include comparison against reference benchmarks
4	复现情况 Reproducibility Status	提交包能跑到什么程度，有什么已知限制 Expected performance of the submitted package, and known limitations
5	开源情况 Open-source Status	仓库地址、许可证、可复用的部分 Repository URL, license, and reusable components

- 格式：PDF 或 PPTX
- 命名：与提交包同前缀，如 **OPENxxx_作品名称_slides.pdf**

- 页数上限：【待定】
- 答辩时间与形式另行通知
- Format: PDF or PPTX
- Naming: Same prefix as the submission archive, e.g.
OPENxxx_SubmissionTitle_slides.pdf
- Page Limit: [To-Be-Confirmed]
- Timing and format of the online defense will be notified separately.

PPT 建议：优先说明问题定义为何成立、发现为何可信，便于评审快速理解评估对象。

PPT Recommendation: Prioritize justifying the validity of problem definition and the credibility of your findings to help judges rapidly understand judging objects.

2. 科学发现与环境定义报告

2. Scientific Finding & Environment Definition Report

报告内容要求：

Report requirements:

1. 科学问题与发现目标

1. Scientific Problem & Discovery Objectives

研究的科学问题是什么？为什么值得研究？希望 Agent 发现什么类型的新规律、新信号、新结构或新假设？**最终声称的核心科学发现（Scientific Claim）是什么？**

What is the target scientific problem? Why is it worth investigating? What new patterns, signals, structures or hypotheses are expected to be discovered by the

Agent? What is the core scientific claim put forward in your work?

2. Environment / Benchmark 定义

2. Environment / Benchmark Definition

- Agent 能观察什么 (Observation)、执行什么操作 (Action)、调用哪些工具 (Tools) ?
- 用了哪些数据、模拟器、实验环境、数据库或外部资源?
- **怎么判断一次发现成立**——评价指标、Verifier、Ground Truth 还是别的验证机制?
- 如果存在隐藏答案、测试集或 Ground Truth, **必须明确哪些信息 Agent 可见 / 不可见**, 避免信息泄漏。
- What Observations can the Agent obtain, what Actions can it perform, and which Tools can it invoke?
- What datasets, simulators, experimental environments, databases or external resources are adopted?
- **How is a valid discovery judged:** via evaluation metrics, Verifier, Ground Truth or other validation mechanisms?
- If hidden answers, test sets or Ground Truth are present, **explicitly define information visible / invisible to the Agent** to prevent information leakage.

3. Agent 如何完成发现

3. Agent Discovery Workflow

总体架构和 workflow: 如何提出假设、调用工具、跑实验或分析数据、修正判断、得到结论。

Overall architecture and workflow: how hypotheses are generated, tools invoked, experiments executed or data analyzed, judgments revised, and final conclusions

reached.

报告要能和提交的 trajectory / log 对应上，让评审能追溯「发现是怎么产生的」。

The report shall align with submitted trajectories / logs so that judges can trace how discoveries are generated.

如果正式运行中存在人工干预、结果筛选、Best-of-N 或多次重跑，必须如实说明。

If manual intervention, result filtering, Best-of-N sampling or multiple re-runs are applied in official executions, such facts must be truthfully documented.

4. 发现结果与验证

4. Discovery Outcomes & Validation

给出核心发现和主要证据，并说明独立验证、重复实验、不同随机种子、对照实验或稳健性检验。

Present core findings and key supporting evidence. Describe independent validation, repeated experiments, tests with different random seeds, control-group experiments and robustness checks.

对随机性强的 Agent，**不要求逐 token 或逐 action 完全一致**，但要说明发现的稳定性、成功率和主要波动。

For Agents exhibiting high stochasticity, **exact token-by-token or action-by-action replication is not required**. Document discovery stability, success rates and major fluctuation patterns.

简要说明当前发现最主要的 **limitation** 和可能的 **false positive**。

Briefly state major **limitations and potential false positive** risks of current findings.

5. 开源资源链接

5. Open-source Resource Links

- 代码仓库链接（GitHub / GitLab / Gitee 或其他公开平台）
- 数据 / Benchmark 链接：自建的，提供 Hugging Face、Zenodo、Figshare 等公开渠道链接，并给出关键构建流程；公开数据要注明来源、版本 / revision 和预处理方式。
- Code repository links (GitHub / GitLab / Gitee or other public platforms)
- Data / Benchmark links: For self-built resources, provide links via public platforms such as Hugging Face, Zenodo or Figshare together with key build-workflow documentation. For public datasets, specify source, version / revision and pre-processing procedures.

三、代码类材料

IV. Code Materials

1. 三类核心复现材料

1. Three Core Reproducibility Materials

开放探索赛题需提交以下三类核心复现材料，**缺一项即为材料不完整**：

The Open Exploration Sub-track requires submission of all three core reproducibility materials. Missing any one will render the submission incomplete.

① **最小可运行探索环境**——不要求功能齐全，但要能装、能跑、能复现你报告里的核心结果。README 要写清环境安装、依赖、配置方法、关键版本和预期输出。

① **Minimal Operable Exploration Environment**: Full feature completeness is not mandatory, but the environment must be installable, runnable, and capable of reproducing core results described in the report. The README shall clearly document environment installation, dependencies, configuration approaches,

key versions and expected outputs.

② **至少一次完整探索日志**——与正式结果对应的完整运行记录。至少能追溯：Agent 输入与 observation、输出与 action、tool call 及返回、环境反馈、关键中间产物、最终结果与指标、必要的错误与重试。**推荐 JSON / JSONL，不要只交截图或录屏。**

② **At Least One Complete Exploration Log:** Full runtime record corresponding to official results. It shall at minimum trace: Agent inputs and Observations, outputs and Actions, tool-call payloads and returns, environment feedback, key intermediate artifacts, final results and metrics, as well as relevant error and retry events. **JSON / JSONL is preferred. Screenshots or screen recordings alone are not acceptable.**

③ **参照系设计**——需证明「发现」并非随机波动带来的偶然结果。参照系可以是随机探索、平凡解或简单基线，并需提交对应结果供评审对比。

③ **Reference Benchmark Design:** Prove that findings are not accidental artifacts caused by random fluctuations. Reference benchmarks may include random exploration runs, trivial solution baselines or simple baselines. Corresponding outputs shall be provided for judge comparison.

2. 仓库的最低要求

2. Minimum Repository Requirements

不强制统一项目结构，但必须满足：

No unified project structure is enforced, yet all below requirements must be satisfied:

1. **README 要让评审知道「怎么跑」**——环境与依赖安装、数据如何获得、用的模型 / API 及关键配置、大致算力与成本、smoke test 怎么跑、核心结果怎么复现、跑成功应得到什么输出。

1. **The README shall enable judges to execute the workflow:** environment and dependency installation, data acquisition, model / API selections and key

configurations, approximate compute resource and cost estimates, smoke test procedures, core-result reproduction steps, and expected successful outputs.

2. **完整 trajectory / log**——至少能追溯：Agent 输入与 observation、输出与 action、tool call 及返回、Environment feedback、关键中间产物、最终结果与指标、必要的错误与重试信息。

2. **Complete trajectories / logs**: At minimum trace Agent inputs & Observations, outputs & Actions, tool-call requests & returns, environment feedback, key intermediate artifacts, final results & metrics, plus necessary error and retry records.

推荐 JSON / JSONL 等机器可解析格式，不要只交截图或终端录屏。模型平台无法导出的 private reasoning 不作要求，但 Agent 实际接收到和实际执行的内容必须完整保存。

Machine-parsable formats such as JSON / JSONL are recommended; screenshots or terminal recordings alone are invalid. Private reasoning that cannot be exported by model platforms are exempt, yet **content actually received and executed by the Agent must be fully preserved**.

3. **可复现配置**——依赖与环境、模型名称与版本、主要 sampling 参数、random seed、数据版本、正式结果对应的 run / config、第三方代码来源及 License。不得上传 API Key、密码或其他 secret。

3. **Reproducible configurations**: Dependencies and environment settings, model names and versions, primary sampling parameters, random seeds, dataset versions, run / config corresponding to official results, third-party code origins and licenses. **Do not upload API keys, passwords or other secrets**.

4. **数据与自建 Benchmark**——公开数据建议提供自动下载 / 预处理脚本；自建 benchmark 需开源关键构建流程并提供可访问的数据或稳定引用。因版权、隐私或许不能公开的部分，要在报告和 README 中说明其对复现的影响。

4. **Datasets & Self-Built Benchmarks**: Provide automatic download / pre-processing scripts for public datasets. For self-built benchmarks, open-source

key construction workflows and supply accessible data or stable references. For components that cannot be released due to copyright, privacy or licensing restrictions, **document their impact on reproducibility within both report and README.**

5. 统一入口——建议提供：

5. **Unified entry-point:** the following are recommended:

```
bash
bash scripts/smoke_test.sh
bash scripts/reproduce_core.sh
```

评审不应手动翻多个 notebook 才能拼出完整流程。

Judges shall not need to manually piece together workflows across multiple notebooks.

结果必须可追溯

Mandatory Traceability Requirement

报告中的主要 Figure、Table、指标或 Scientific Claim，要能追溯到：

Major Figure, Table, metric or Scientific Claim in the report shall be traceable through the chain below:

代码版本 → 配置 → 数据版本 → Agent trajectory → 结果文件
Code Version → Configurations → Dataset Version → Agent Trajectory →
Output Files

如果最终结果来自多次运行或人工筛选，**必须明确记录总运行次数、筛选规则、正式采用的 run。**

If final results originate from multiple runs or manual filtering, **explicitly document total run counts, filtering criteria and the exact run officially**

adopted.

四、其他建议和说明

IV. Additional Notes & Recommendations

1. 复赛评审权重

1. Semi-Final Judging Weighting

开放探索赛题按以下四个维度评审：

The Open Exploration Sub-track adopts four judging dimensions as follows:

评审维度 Judging Dimension	权重 Weighting	关键得分点 Key Scoring Criteria
问题定义与环境设计质量 Problem Definition & Environment Design Quality	45%	研究问题真实且边界清晰；探索环境抓住问题本质；固定部分、可探索部分和反馈机制明确 Research problem is authentic with well-defined boundaries; exploration environment captures the essence of the problem; fixed components, explorable components and feedback mechanisms are clearly defined.
探索过程与科学/研究信号 Exploration Process &	35%	产生有意义的正向发现、异常、反例、可解释的负结果，或对问题定义的有效修正 Generation of meaningful positive

Scientific / Research Signals		findings, anomalies, counter-examples, interpretable negative results, or effective revisions to problem definition.
可检查性与可延续性 Inspectability & Sustainability	15%	提供最小参照系、运行日志、复现说明和后续研究路径，成果可沉淀为可接续的问题包或环境包 Provide minimal reference benchmarks, runtime logs, reproducibility instructions and pathways for follow-up research. Deliverables can be packaged as reusable problem sets or environment packages for further work.
开源贡献 Open-source Contribution	5%	代码、探索环境或数据处理管线的开源质量、可复用性 Open-source quality and reusability of code, exploration environments or data-processing pipelines.

评审口径：问题定义与环境设计占 45%。开放探索赛题不仅关注发现本身，也关注问题定义是否合理、环境是否可运行、评估口径是否清楚。**负结果可以提交，但需说明失败原因及其价值。**

Judging Remarks: Problem Definition & Environment Design accounts for 45 % of total weight. The Open Exploration Sub-track evaluates not only final discoveries themselves, but also the rationality of problem definition, environment operability and clarity of evaluation methodology. **Negative-result submissions are acceptable provided failure causes and their research value are**

explained.

2. 这些情况不接受

2. Disqualifying Submission Scenarios

- 只交结论，没有对应的探索日志
- **没有参照系**，无法排除「发现」来自随机波动
- 只交截图或录屏当运行记录
- 仓库在评审期间持续变动，没有固定 tag / commit
- 存在人工干预或多次重跑但未说明
- 报告中的结果与提交的日志对不上
- 包内含有 API Key、密码或其他 secret
- Conclusions submitted without corresponding exploration logs
- **Missing reference-benchmarks**, making it impossible to rule out findings arising purely from random fluctuations
- Screenshots or screen recordings submitted in place of runtime logs
- Repository continuously modified throughout evaluation without fixed tag / commit
- Manual intervention or multiple re-runs applied without proper disclosure
- Reported results inconsistent with submitted logs
- Archive contains API keys, passwords or other secrets

3. 提交前自检

3. Pre-Submission Self-Check Checklist

☐ **三件套齐全**：可运行探索环境、完整探索日志、参照系设计

- ☐ 报告清楚定义了科学问题、Environment 和核心发现
- ☐ 报告包含代码仓库以及必要的数据 / Benchmark 开源链接
- ☐ 仓库 README 可以指导第三方完成安装和运行
- ☐ **Complete three-component set:** operable exploration environment, full exploration logs, reference-benchmark design
- ☐ Scientific problem, Environment and core findings are clearly defined in the report
- ☐ Code-repository links and required open-source links for data / Benchmarks are included in the report
- ☐ Repository README enables third-party installation and execution
- ☐ 提供一键 smoke test，并尽可能提供核心结果的一键复现
- ☐ 正式结果对应的**完整 Agent trajectory / log 已提交**
- ☐ 关键配置、模型版本、数据版本、随机种子已记录
- ☐ 人工干预、多次运行、结果筛选方式已如实说明
- ☐ One-click smoke test is available; one-click reproduction for core results is provided to the extent feasible
- ☐ **Complete Agent trajectories / logs submitted** corresponding to official results
- ☐ Key configurations, model versions, dataset versions and random-seed values are documented
- ☐ Manual intervention, multiple runs and result-filtering rules are truthfully stated
- ☐ 最终科学结果能从报告一路追溯到代码、轨迹和结果文件
- ☐ 仓库已固定 release / tag / commit hash
- ☐ 包内不含 API Key、密码或未授权第三方数据
- ☐ Final scientific outcomes can be traced from report back to code, trajectories

and output files

☐ Repository release / tag / commit-hash has been frozen

☐ Submission archive contains no API keys, passwords or unauthorized third-party data